

Comprehensive AI Developer Course

(Machine Learning, Deep Learning, Computer Vision & LangChain)

Module 1: Foundations of Machine Learning (ML)

Introduction to ML

- What is Machine Learning?
- Types: Supervised, Unsupervised, Reinforcement
- ML vs AI vs Data Science
- Real-world applications

Supervised Learning

- Linear & Logistic Regression
- Decision Trees & Random Forests
- K-Nearest Neighbors
- Support Vector Machines (SVM)
- Evaluation Metrics: Accuracy, Precision, Recall, F1-Score

Unsupervised Learning

- Clustering (K-Means, DBSCAN, Hierarchical)
- Dimensionality Reduction (PCA, t-SNE)

Model Tuning

- Cross-validation
- Hyperparameter tuning (GridSearchCV, RandomizedSearchCV)
- Feature scaling and normalization
- Pipeline automation in scikit-learn

Tools & Libraries

- NumPy, Pandas, Matplotlib, Seaborn
- scikit-learn, XGBoost, LightGBM

Module 2: Deep Learning

Fundamentals of Neural Networks

- Perceptron, activation functions
- Feedforward Neural Networks
- Backpropagation and loss functions

Deep Learning Frameworks

- TensorFlow 2.x & Keras API
- PyTorch basics and comparison

Advanced Neural Networks

- Convolutional Neural Networks (CNNs)
- Recurrent Neural Networks (RNNs), LSTMs, GRUs
- Transfer learning with pretrained models

Training Deep Models

- Optimizers: SGD, Adam, RMSProp
- Regularization: Dropout, L2, Early Stopping
- Batch Normalization
- GPU acceleration with CUDA

Projects

- MNIST digit classifier
- Sentiment analysis with LSTM
- Image classification with ResNet/VGG

Module 3: Computer Vision (CV)

Introduction to CV

- Image basics (pixels, color spaces)
- Reading and processing images with OpenCV

Image Processing Techniques

- Filtering, thresholding, edge detection
- Morphological operations
- Contours and shape detection

Deep Learning for CV

- CNN architecture in depth
- Data augmentation techniques
- Object Detection: YOLO, SSD, Faster R-CNN
- Image segmentation: U-Net, Mask R-CNN

Advanced Applications

- Face detection & recognition

- OCR with Tesseract & CRNN
- Pose estimation and tracking

Tools & Libraries

- OpenCV, Pillow, Albumentations
- torchvision, detectron2, YOLOv8

Module 4: LangChain & LLMs

Introduction to Large Language Models

- What is a LLM?
- Overview of GPT, Claude, LLaMA, Gemini
- Prompt engineering basics

Introduction to LangChain

- What is LangChain and why use it?
- Components of LangChain: Chains, Agents, Tools
- Setting up LangChain with OpenAI and HuggingFace

LangChain Core Concepts

- Document loading & indexing
- Memory and conversation tracking
- Retrieval-Augmented Generation (RAG)
- Prompt templates and chains

LangChain + Vector Databases

- Introduction to vector embeddings
- FAISS, ChromaDB, Pinecone
- Building a semantic search engine

LangChain Projects

- AI PDF Q&A bot
- Conversational chatbot with memory
- LLM-powered document summarizer
- Code generation assistant

Tools & Ecosystem

- LangChain, LlamaIndex
- OpenAI, Cohere, HuggingFace

- Streamlit/Gradio for interface

Capstone Project (Choose One or Combine)

- Smart Search Engine using LangChain + vector DB
- AI Image Classifier with Deep Learning and CV
- Multimodal App combining NLP + Vision
- AI Tutor Bot that answers from course PDFs